

Elliptical slice sampling

A two-variable example and MATLAB implementation

Econometrics 32706

The target distribution

Suppose a latent vector has a Gaussian prior and a likelihood:

$$f \sim \mathcal{N}(0, K), \quad p(f \mid y) \propto \mathcal{N}(f; 0, K)L(f).$$

- Elliptical slice sampling (ESS) updates the whole vector f .
- We need a draw from $\mathcal{N}(0, K)$ and a function that returns $\log L(f)$.
- The likelihood can be non-Gaussian. The prior covariance K is held fixed during this update.

A proposal on a Gaussian ellipse

From the current state f , draw an independent $\nu \sim \mathcal{N}(0, K)$. For any angle θ , define

$$f(\theta) = f \cos \theta + \nu \sin \theta.$$

Angle	Candidate	Meaning
0	f	current state
$\pi/2$	ν	independent prior draw
π	$-f$	opposite point on the ellipse

Rotating the pair (f, ν) preserves their joint Gaussian prior. The likelihood determines which angles are eligible.

Why the rotation preserves the Gaussian prior

Under the prior, f and ν are independent draws from $\mathcal{N}(0, K)$. Rotate *both* vectors:

$$\begin{aligned}f_{\theta} &= f \cos \theta + \nu \sin \theta, \\ \nu_{\theta} &= -f \sin \theta + \nu \cos \theta.\end{aligned}$$

The new first vector still has covariance K :

$$\text{Var}(f_{\theta}) = (\cos^2 \theta + \sin^2 \theta)K = K, \quad \text{Cov}(f_{\theta}, \nu_{\theta}) = 0.$$

The rotated pair therefore has the same joint Gaussian prior. In ESS, the likelihood slice then selects an eligible angle, bringing the observed data into the update.

What orthogonal means here

The coefficients in the two rotation equations form the matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \quad R(\theta)^\top R(\theta) = I.$$

Its columns have length one and are perpendicular. Therefore, mixing the pair (f, ν) does not change its combined squared length:

$$\|f_\theta\|^2 + \|\nu_\theta\|^2 = \|f\|^2 + \|\nu\|^2.$$

At $\theta = \pi/2$, the pair becomes $(f_\theta, \nu_\theta) = (\nu, -f)$: the vectors swap roles, with one sign changed.

One ESS update

1. Draw $\nu \sim \mathcal{N}(0, K)$ and $u \sim U(0, 1)$. Set the log slice level $\ell_\star = \log L(f) + \log u$.
2. Draw an initial angle $\theta \sim U(0, 2\pi)$. Set $[\theta_{\min}, \theta_{\max}] = [\theta - 2\pi, \theta]$.
3. Test $f(\theta) = f \cos \theta + \nu \sin \theta$. If $\log L(f(\theta)) > \ell_\star$, return $f(\theta)$.
4. On rejection, set $\theta_{\min} = \theta$ if $\theta < 0$; otherwise set $\theta_{\max} = \theta$. Draw another θ uniformly from the remaining bracket and return to step 3.

Keep the same ν and $\ell_\star$ throughout this update.

The initial bracket contains zero

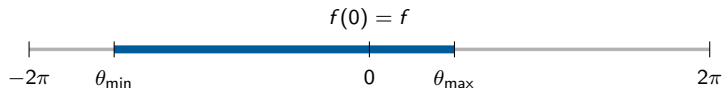
The ellipse repeats every 2π :

$$f(\theta) = f \cos \theta + \nu \sin \theta, \quad f(2\pi k) = f \quad (k \in \mathbb{Z}).$$

For an initial draw $0 < \theta < 2\pi$, set

$$\theta_{\min} = \theta - 2\pi, \quad \theta_{\max} = \theta.$$

Thus $\theta_{\min} < 0 < \theta_{\max}$, and the bracket spans exactly one full turn.



Why the initial endpoints agree

At initialization, $\theta_{\min} = \theta_{\max} - 2\pi$. Periodicity gives

$$\begin{aligned} f(\theta_{\min}) &= f \cos(\theta_{\max} - 2\pi) + \nu \sin(\theta_{\max} - 2\pi) \\ &= f \cos \theta_{\max} + \nu \sin \theta_{\max} = f(\theta_{\max}). \end{aligned}$$

This equality holds for every f , ν , and initial angle. It says that the endpoints are the *same point* on the ellipse; it does not say that the point is ν or f .

Separately, $f(0) = f$. Keeping 0 in the bracket preserves a known acceptable angle while the sampler searches.

The drawn angle sets both endpoints

At initialization, draw $\theta \in (0, 2\pi)$ and set

$$[\theta_{\min}, \theta_{\max}] = [\theta - 2\pi, \theta].$$

These two draws give different brackets and endpoint values:

Drawn θ	Bracket $[\theta_{\min}, \theta_{\max}]$	Value at both endpoints
$\pi/2$	$[-3\pi/2, \pi/2]$	ν
$\pi/3$	$[-5\pi/3, \pi/3]$	$\frac{1}{2}f + \frac{\sqrt{3}}{2}\nu$

The endpoints agree because they are 2π apart. Their shared value depends on the drawn angle. The current state f is at angle 0 in both brackets.

Why shrinking leaves acceptable angles

Write $g(\theta) = \log L(f(\theta))$. Since $u \in (0, 1)$, the slice level is strictly below the value at zero:

$$\ell_{\star} = g(0) + \log u < g(0).$$

If g is continuous at zero, some $\varepsilon > 0$ satisfies

$$|\theta| < \varepsilon \implies g(\theta) > \ell_{\star}.$$

Thus a small interval around zero is acceptable. A rejected angle lies outside it. Shrinking the endpoint on that angle's side of zero keeps an acceptable interval in the bracket.



Green: accepted near zero. Blue: remaining bracket. Orange: discarded. The case $\theta < 0$ is symmetric.

This local argument needs continuity of $\log L(f(\theta))$ at zero; continuity of $f(\theta)$ alone is insufficient.

A two-variable example

Let the variables have a correlated prior. Observe the first variable with Gaussian noise:

$$K = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}, \quad y = f_1 + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.35^2), \quad y = 1.$$

The log-likelihood, up to a constant, is

$$\log L(f) = -\frac{(1 - f_1)^2}{2(0.35)^2}.$$

In this linear-Gaussian case, the exact posterior is available:

$$\mathbb{E}[f \mid y] \approx \begin{pmatrix} 0.891 \\ 0.713 \end{pmatrix}, \quad \text{Var}(f \mid y) \approx \begin{pmatrix} 0.109 & 0.087 \\ 0.087 & 0.430 \end{pmatrix}.$$

The observation moves f_2 as well because the prior correlates it with f_1 .

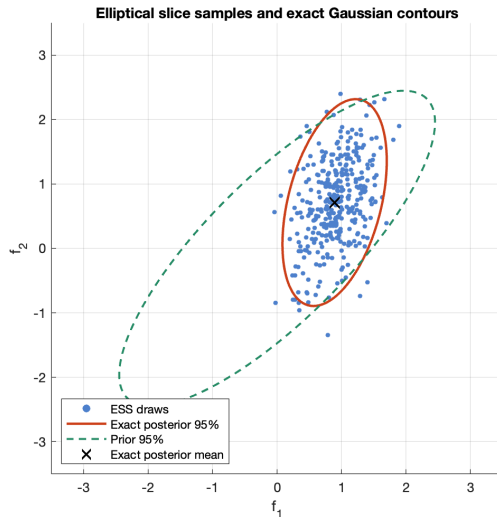
MATLAB demonstration

`elliptical_slice_step.m` implements one update. `demo_ess.m` runs the example and compares moments.

```
K = [1, .8; .8, 1];  
cholK = chol(K, 'lower');  
loglike = @(f) -.5*((1 - f(1))/ .35)^2;  
f = zeros(2,1);  
for t = 1:7000  
    f = elliptical_slice_step(f, cholK, loglike);  
end
```

The demo discards 1,000 initial draws, retains 6,000, prints sample and exact moments, and plots the draws with exact contours.

Samples follow the exact posterior



What the comparison checks

- The sample mean should be near $(0.891, 0.713)^\top$.
- The sample cloud should follow the exact posterior contour, which is narrower than the prior in the f_1 direction.
- This example has an exact posterior only because the likelihood is Gaussian. The same ESS code can use a different log-likelihood.

Source: I. Murray, R. P. Adams and D. J. C. MacKay (2010), *Elliptical slice sampling*, AISTATS, pp. 541–548. <https://proceedings.mlr.press/v9/murray10a.html>